THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS UNIT TEST SET C

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Vertically opposite angles are always: 1

(a) complementary **(b)** unequal **(c)** equal **(d)** supplementary

2. The complement of 38° is: 1

(a) 52° **(b)** 128° **(c)** 38° **(d)** 142°

3. Two lines intersect and one angle is 55°. The vertically opposite angle is: 1

(a) 250° **(b)** 125° **(c)** 35° **(d)** 55°

4. The product 13 × 11 is: 1

(a) odd **(b)** even **(c)** neither **(d)** both

5. The sum of two odd numbers is always: 1

(a) a multiple of 3 **(b)** even **(c)** odd **(d)** prime

6. If 6 × B ends in 4, one value of B is: 1

(a) 3 **(b)** 7 **(c)** 4 **(d)** 5

7. Two angles of a triangle are 35° and 75°. The third angle is: 1

(a) 70° **(b)** 110° **(c)** 145° **(d)** 55°

8. The three altitudes of a triangle meet at a point called: 1

(a) incentre **(b)** orthocentre **(c)** circumcentre **(d)** centroid

9. Sides 3 cm, 4 cm and 5 cm form a triangle? 1

(a) Only if equal **(b)** Cannot say **(c)** Yes **(d)** No

10. 3/4 ÷ 1/2 = 1

(a) 3/2 **(b)** 3/8 **(c)** 1/4 **(d)** 2/3

11. Dividing a number by a proper fraction gives a result that is: 1

(a) always 1 **(b)** smaller than the number

(c) greater than the number **(d)** equal to the number

12. 5/8 of 72 is: 1

(a) 48 **(b)** 45 **(c)** 581/8 **(d)** 576/5

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

13. How can you check if two lines are parallel using a transversal? 2

14. Is 3,654 divisible by 9? Show. 2

15. Name the types of triangles based on angles. 2

16. How many pieces of 1/8 m can be cut from 3 m ribbon? 2

17. Why is dividing by 1/2 the same as multiplying by 2? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. Two lines intersect so that one angle is 4x and the adjacent angle is 5x. Find all four angles. 3

19. Solve the cryptarithm: A B + A B = B C where A, B, C digits. 3

20. The perimeter of an isosceles triangle is 36 cm and the base 10 cm. Find the equal sides. 3

21. A recipe uses 3/4 cup of flour for one batch. How much for 2 1/2 batches? 3

22. Is the product 13 × 14 × 15 odd or even? Explain. 3

OR

Write the grid pattern for multiples of 9 up to 99 and state one observation. 3

23. Draw the idea of a median and show that a triangle has three. 3

OR

How do you construct an altitude of a triangle? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

24. (i) Write the first 10 terms of the Virahāṅka sequence. 5
(ii) Show each rule step.
(iii) Which term is 144?

OR

Case study: Meena builds a 3 × 3 magic square with centre 5 and 1–9. 5
(i) What is the magic sum?
(ii) Which numbers go at corners?
(iii) Give one full square.
(iv) Check one diagonal.

25. (i) Multiply 3 1/3 by 2 1/4. 5
(ii) Divide the answer by 5/6.
(iii) Write each as a mixed number.

OR

(i) A cyclist covers 4 1/2 km in 3/4 hour. Find speed. 5
(ii) Distance in 2 1/2 h.
(iii) Time for 27 km.

END OF QUESTION PAPER — Total: 50 Marks